Team #21

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General Idea: 4 days of intraday data, 10,000 ticks of it, no transaction fees, evaluated on PnL out of sample + write up -> due Nov 1st, 11:59 pm

Python and Csv files

Basic idea -> use statistical methods to find patterns in the data without overfitting, and aggressively test on different out of sample conditions

No transaction fees, so trade as much as you possibly can, if it is positive expectancy at all, take the trade dw about fees

Use Claude/CODEX/models to generate synthetic out of sample data perhaps similar to what they would give us and then check how the strategy performs on that

Make sure to test point in time, and eliminate any look ahead bias, strictly, also being careful with inputs to our own model

How can we generate good synthetic data from claude or any model and how cna we evaluate if the data is realistic or not,

Potential starting point reading some academic papers for this sort of a problem, and seeing how monte carlo or other numerical methods can be used to find a robust strategy

Start with some simple model on the data like linear regression or smth, to find, what is making up the potential factors

Are there any factors that we are looking at or is this purely based on teh price (considering it is only 4 days, its probably just price only)

What strategies could we use: statistical arbitrage (im pretty sure this is what they expect), trend following/momentum, mean reversion, are there pairs we can potentially trade, like cross-asset relationships to trade

Grid optimization for parameters potentially could be used (this is what we did last year)